Guidelines for Management of Meningoencephalitis
Version 1.2

Purpose: To guide clinicians on the initial management approach to the pediatric patients with suspected acute bacterial meningitis with or without encephalitis

Scope: Department of pediatrics, Department Pediatric Emergency and Department of Pediatric Critical Care

Definitions, Signs and Symptoms:

Meningitis:

Meningitis is an inflammation of the meninges surrounding the brain and spinal cord.

Unlike adult patients who usually present with fever, headache, meningeal signs ± photophobia± altered mental status; children usually present with fever, poor oral intake, vomiting, and lethargy or irritability ± seizure. About 90% of bacterial meningitis cases are caused by *Streptococcus pneumoniae*.

Encephalitis:

Inflammation of the cerebral cortex. The patient presents with fever, headache & altered mental status, behavioral, personality and speech problem, obtundation and coma. Encephalitis most commonly caused by viruses.

Antimicrobial therapy of meningitis and encephalitis based on pathogen

CNS Infection	Age	Organisms	Antimicrobial Agent
Meningitis	Neonate	Group B *Streptococcus*, gram negative enteric bacilli (*E.coli*), rarely *Listeria monocytogenes*	Cefotaxime ± Ampicillin
Meningitis	1-3 months	Includes organisms usually seen in neonates or older children	Cefotaxime+ Vancomycin ± Ampicillin
Meningitis	Older Children	*S.pneumonia, N. meningtitides, H.Influenzae*	Ceftriaxone + Vancomycin
Encephalitis	Pediatrics	Herpes simplex	IV Acyclovir 20mg/kg/dose every 8hrs and for 21 days

Antimicrobial therapy of other CNS Infections

CNS Infection	Organisms	Antimicrobial Agent
Shunt Infection	*Coagulase-negative, Staphylococcus, coliform bacilli, Pseudomonas*	Vancomycin + Ceftriaxone Or Vancomycin + Ceftazidime Or Cefepime or Meropenem
Subdural empyema	Most of the time its extension of sinusitis or otitis media	Same as for primary brain abscess Surgical emergency and must be drained.
Brain abscess Consult Infectious diseases and neurosurgical consult for potential drainage	*Strept. spp. Bacteroides*, Enterobacteriaceae, *S.milleri*, *S.aureus*, Rare: *Nocardia*, *Listeria* *Toxoplasma gondii* in HIV patient	Ceftriaxone + Vancomycin + Metronidazole • Use initially Vancomycin to cover MRSA or beta lactam resistant *S.pneumoniae* then adjusts according to culture results • Use Ceftazidime instead of Ceftriaxone if secondary to chronic otitis externa • Meropenem may be used in situation of antibiotic resistant (ESBL) • *S. aureus* more likely if h/o endocarditis and positive blood culture.

I. Meningitis

How to differentiate between bacterial and viral meningitis?

Performing lumber puncture and sending CSF microscopy, biochemistry, culture and Viral PCR. If CSF culture and viral PCR is negative, no definitive test can differentiate bacterial meningitis from nonbacterial meningitis. High WBC counts, positive gram stain results, low glucose concentrations, and high protein level are neither sensitive nor specific to reliably distinguish bacterial from nonbacterial meningitis.

CSF bacterial PCR: Could be considered in case of negative culture or undertreated or partially treated meningitis.

Indication of CT head prior to lumber puncture:

- History of CNS disease (e.g., mass lesion, stroke and focal infection), an immunocompromised state (e.g., that due to HIV infection or AIDS, immunosuppressive therapy or transplantation)
- History of prolong seizure
- Certain specific abnormal neurologic findings (e.g. an abnormal level of consciousness, an inability to answer 2 consecutive questions correctly or to follow 2 consecutive commands, gaze palsy, abnormal visual fields, facial palsy, arm drift, leg drift, abnormal language).
- Signs of increase intracranial pressure like papilledema. Treat high intracranial pressure then to do CT and LP if possible

Lumbar puncture contraindications:

- Skin infection near the site of the lumbar puncture
- Suspicion of increased intracranial pressure due to a cerebral mass.
- Uncorrected coagulopathy
- Acute spinal cord trauma
- Markedly reduced GCS
- Cardiovascular compromise

Indications for repeating lumbar puncture in patients with meningitis:

- If the patient doesn't response clinically after 48h of appropriate antimicrobial therapy
- Neonates with gram-negative bacilli meningitis should undergo repeated lumbar punctures to document CSF sterilization, as the duration of antimicrobial therapy is determined, in part, by the result
- In patients with CSF shunt infections, the presence of drainage catheter after shunt removal allows for monitoring of CSF parameters to ensure that the infection is responding to appropriate antimicrobial therapy and drainage

Antimicrobial therapy doses for children:

- Vancomycin: 15 mg/kg/dose q6h for normal renal function test.
- Ceftriaxone: 50 mg/kg/dose q12 or
- Cefotaxime: 50 mg/kg/dose q6h.
- Ampicillin: 100mg/kg/dose q6h.
- Penicillin G >28 days: 300,000 units/kg/day q4h.
- Acyclovir
    - <3 months: 20 mg/kg/dose q8h.
    - >3 month: 15mg/kg/dose q8h.

Duration of antimicrobial therapy for bacterial meningitis based on isolated microorganism

Micro organism	**Duration of therapy in days**
Neisseria meningitides	7
Haemophilus influenza	10
Streptococcus pneumoniae	10-14
Streptococcus agalactiae	14-21
Staph. aureus	14
Aerobic gram-negative bacilli	21
Listeria monocytogenes	21

Role of dexamethasone therapy in-patients with bacterial meningitis?

- ***Neonates:*** Steroids aren't encouraged in neonates due to feasible results on neurodevelopment
- ***Infants and children:*** At present, there are mixed data to make a recommendation on the use of adjunctive dexamethasone in neonates with bacterial meningitis.

NB: The first dose of IV dexamethasone should be given within 4 hours of initial antibiotics use and not more than 12 hours after starting antibiotics

Criteria for Outpatient Antimicrobial Therapy (OPAT) in patients with bacterial meningitis?

- Received at least ≥ 6 days of inpatient antimicrobial therapy
- Afebrile for at least 24 - 48 h prior to discharge for outpatient therapy
- No significant neurological deficit, focal finding or seizure
- Clinical stability or improving condition
- Able to fluids by mouth
- Reliable intravenous access
- Established management plan
- Safe environment with easy access to the medical facility

II. Encephalitis

Diagnosis requires MRI - sensitive and specific

- **Empirical Therapy:** Acyclovir should be initiated in all patients with suspected encephalitis such as those with alter mental status, acute cognitive dysfunction, behavioral changes, focal neurologic signs, and prolonged seizures. However, in most cases, there is some concomitant meningeal inflammation.
- **Specific Therapy**

  Viruses

  - Herpes simplex or varicella-zoster virus: acyclovir is recommended
  - Epstein Barr virus, enteroviruses & mumps: no antivirals recommended
  - Human herpesvirus 6: ganciclovir or foscarnet in immunocompromised patients

- Measles: ribavirin can be considered, intrathecal ribavirin can be considered in patients with subacute sclerosing panencephalitis

III. Notification:

- All patients with suspected or confirmed *N meningitidis*, *pneumococcal* meningitis and *H.Influenzae* should be notified to the Ministry of Health.
- All cases of acute encephalitis syndrome should be notified to the Ministry of Health.

IV. Follow up:

- Children with encephalitis or bacterial meningitis need to have a formal audiology assessment 6–8 weeks after discharge (in advance if concerns)
- Neurodevelopmental progress ought to be monitored in outpatients
- Consider investigating for complement deficiency if the child has had >1 episode of meningococcal disease

References:

Title of book/ journal/ articles/ Website	**Author**	**Year of publication**	**Page**
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Review / Revision History

	Revision History		
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